SalesGenie AI Platform

Software Requirements Specification (SRS)

AI Sales Assistant & Lead Intelligence Platform

Document Control

Field	Detail
Document	Software Requirements Specification (SRS)
Product	SalesGenie AI Platform
Standard	IEEE Std 830-1998 (Clauses 3.1–3.3, 5.2.3, 5.3.6)
Requirements Syntax	EARS — When [event/stimulus], the system shall [response]
Parties	Customer (funding & acceptance authority) and Supplier (engineering & delivery)
Technology Stack	FastAPI, Streamlit, LangGraph, PostgreSQL, Gemini / OpenAI LLMs

Prepared in accordance with IEEE Std 830-1998

Table of Contents

1. Introduction & Scope 3

1.1 Strategic Context & Purpose 3

1.2 Product Scope & Business Objectives 3

1.3 Definitions, Acronyms & Reference Standards 4

2. Overall Description & Enterprise Architecture 5

2.1 Strategic Architecture Context 5

2.2 Product Perspective & Architectural Layers 5

2.3 User Classes & Operational Characteristics 6

2.4 Database Schema & Data Model Architecture 6

2.5 Operational Constraints & Dependencies 7

3. Core Functional Requirements (Modules 1–6) 8

3.1 Strategic Functional Overview 8

3.2 Module 1: Lead Management & Prospect Database 8

3.3 Module 2: Lead Intelligence & Company Analysis 9

3.4 Module 3: AI Outreach Generation 9

3.5 Module 4: Lead Scoring & Recommendation Engine 9

3.6 Module 5: Conversation Intelligence & CRM Integration 10

3.7 Module 6: Dashboard & Sales Analytics 10

4. Key Non-Functional Requirements & Risk Management 11

4.1 Strategic Quality & Reliability Context 11

4.2 Performance & System Attributes 11

4.3 Risk Management & Failure Modes (FMEA Matrix) 11

5. Milestone Build Plan & Evaluation Metrics 12

5.1 Strategic Execution Context 12

5.2 Week-Wise Implementation Roadmap 12

5.3 Milestone Evaluation Criteria & Acceptance Benchmarks 13

1. Introduction & Scope

1.1 Strategic Context & Purpose

In modern enterprise sales operations, revenue generation teams experience systemic operational friction characterized by time-consuming prospect research, fragmented account intelligence, inconsistent outreach messaging, and suboptimal lead qualification. These inefficiencies directly reduce conversion rates, extend sales cycle duration, and constrain total pipeline throughput. The SalesGenie AI Platform addresses these operational challenges by delivering an enterprise-grade AI Sales Assistant and Lead Intelligence Platform. By deploying Large Language Models (LLMs), agentic multi-agent orchestration workflows, automated CRM synchronization, and predictive revenue analytics, SalesGenie AI automates the transformation of raw prospect data into actionable, high-converting revenue workflows.

The primary purpose of this Software Requirements Specification (SRS) is to establish an authoritative, unambiguous, and contractually binding technical baseline between the software **Customer** (the enterprise entity funding the platform development and authorizing acceptance criteria) and the software **Supplier** (the engineering organization responsible for building and delivering the platform), in direct accordance with IEEE Std 830-1998 (IEEE Recommended Practice for Software Requirements Specifications, Clauses 3.1–3.3). This document specifies system boundaries, functional specifications, architectural topologies, and verifiable quality attributes to eliminate development ambiguity, minimize rework, and provide a single source of truth for verification and validation across the product lifecycle.

Target Audience Usage Matrix

Role	System Usage & Primary Responsibilities
Software Developers	Guide microservice design, LangGraph multi-agent orchestration architecture, API specifications, and PostgreSQL schemas.
Testers / QA Engineers	Derive automated test suites, validation plans, and quantitative acceptance benchmarks for system sign-off.
RevOps & Analysts	Validate business logic, scoring factor weights, workflow automation rules, and sales performance analytics.
Project Managers	Monitor milestone delivery schedules, track dependencies, and verify requirements traceability matrices.
Business Stakeholders	Review functional bounds, authorize SLAs, and sign off on contractual operational acceptance criteria.

1.2 Product Scope & Business Objectives

SalesGenie AI consolidates prospect management, automated company profiling, predictive lead scoring, personalized cold outreach generation, conversation intelligence, and CRM workflow synchronization into a unified, agentic enterprise platform.

Business Objective	Technical Deliverable / Platform Capability
Automate Prospect Research & Qualification	Company Research Engine & Lead Intelligence Generator utilizing LLM-driven web/data extraction.
Improve Outreach Engagement & Conversion	AI Email Generator & Engagement Strategy Engine outputting personalized multi-channel outreach.
Prioritize High-Value Pipeline Opportunities	Predictive Lead Scoring Engine delivering conversion probability percentages (0–100%) and lead buckets.
Streamline Post-Interaction Operations	Conversation Intelligence Engine extracting structured summaries, action items, and follow-up tasks.

Business Objective	Technical Deliverable / Platform Capability
Ensure Enterprise Workflow Integration	FastAPI Integration Layer executing bidirectional sync with Salesforce, HubSpot, and communication tools.
Provide Executive Revenue Visibility	Streamlit Sales Analytics Dashboard displaying pipeline metrics, conversion rates, and forecasts.

The functional scope of SalesGenie AI explicitly encompasses automated multi-channel lead ingestion, company profile intelligence, automated scoring, draft email generation, meeting transcript summarization, action item extraction, and CRM sync logging. Operational boundaries explicitly exclude hardware-level telecommunication routing, native internal CRM primary data hosting (relying on external API hooks to third-party CRMs), and direct financial payment gateway processing. By automating lead analysis and engagement workflows, the platform transforms manual sales research into streamlined agentic operations.

1.3 Definitions, Acronyms & Reference Standards

Term / Acronym	Type	Definition & Context
IEEE Std 830-1998	Reference Standard	IEEE Recommended Practice for Software Requirements Specifications, defining criteria for clear, verifiable, and traceable software requirements.
Customer	Contractual Role	The person or organization paying for the product and specifying acceptance criteria (IEEE Std 830-1998 Clause 3.2).
Supplier	Contractual Role	The organization responsible for engineering, building, and delivering the software product (IEEE Std 830-1998 Clause 3.3).
LLM	Technology	Large Language Model (specifically Gemini / OpenAI architectures) deployed for natural language understanding, generation, and reasoning.
EARS	Requirements Syntax	Easy Approach to Requirements Syntax; a structured formatting standard requiring the form: *When [event/stimulus], the system shall [response].*
FMEA	Analytical Method	Failure Modes and Effects Analysis; a systematic risk mitigation technique evaluating failure mechanisms, causes, and operational impact.
RPN	Risk Metric	Risk Priority Number: RPN = Severity (S) × Occurrence (O) × Detection (D).
CRM	Domain Platform	Customer Relationship Management system (e.g., Salesforce, HubSpot) operating as the enterprise system of record.
RBAC	Security Control	Role-Based Access Control enforcing system access privileges based on assigned user enterprise roles.
FastAPI & Streamlit	Core Frameworks	Modern asynchronous Python web microservice framework (FastAPI) and interactive UI presentation framework (Streamlit).

Having established the foundational scope, contractual definitions, and reference terminology, Section 2 details the multi-layered enterprise architecture, cross-cutting security framework, user profiles, and underlying database models.

2. Overall Description & Enterprise Architecture

2.1 Strategic Architecture Context

To ensure strict modularity, high operational availability, and resilient multi-agent AI orchestration, SalesGenie AI is engineered using a multi-tiered enterprise architecture. This design cleanly isolates front-end presentation services, microservice API routing, security controls, agentic decision logic, third-party integrations, and persistent database stores, preventing cascading failures and enabling independent scaling across system layers.

2.2 Product Perspective & Architectural Layers

The platform comprises eight core operational layers augmented by a vertical cross-cutting Security & RBAC layer spanning the API and Sales Intelligence boundaries, as illustrated in Figure 1.

1 USERS LAYER	Sales Reps | BDRs | Sales Managers | RevOps Teams	
2 PRESENTATION LAYER	Streamlit Dashboard | Intelligence Portal | Analytics	
3 API LAYER	FastAPI Backend Services	**SECURITY & RBAC Authentication Authorization Role-Based Access Control**
4 SALES INTELLIGENCE LAYER	Lead Management | Research Engine | Scoring Engine	
5 AI LAYER	Outreach Generator | Conversation Intel | Recommendations	
6 AGENTIC AI LAYER	LangGraph Multi-Agent Orchestration (Research, Qualification, Outreach, CRM Update, Follow-Up Agents via Gemini / OpenAI)	
7 INTEGRATION LAYER	Salesforce | HubSpot | LinkedIn | Email | Zoom | Teams	
8 DATA LAYER	PostgreSQL Database | Lead Repository | Activity Store	

Figure 1 — SalesGenie AI architectural topology

Layer Descriptions

- **1. Users Layer:** End-user access roles, including Sales Representatives, Business Development Teams, Sales Managers, and Revenue Operations (RevOps) Teams.
- **2. Presentation Layer:** Front-end interactive interfaces built on Streamlit, comprising the Sales Intelligence Portal, Outreach Workspace, Meeting Summarization View, and Executive Analytics Dashboard.
- **3. API Layer:** Asynchronous REST API layer built on FastAPI, serving as the microservice interface bus for system requests, webhooks, and core platform capabilities.
- **Security & RBAC Layer (cross-cutting):** A vertical security module spanning the API Layer and Sales Intelligence Layer, providing identity authentication, token-based authorization, and Role-Based Access Control across all service endpoints and data access operations.
- **4. Sales Intelligence Layer:** Core domain services including the Lead Management Engine, Prospect Database Manager, Company Research Engine, Lead Intelligence Generator, and Lead Scoring Engine.
- **5. AI Layer:** Outreach Generation Engine, Conversation Intelligence Engine, Follow-Up Recommendation Engine, and Sales Insights Generator.

- **6. Agentic AI Layer:** Multi-agent orchestration framework powered by LangGraph, coordinating specialized execution agents (Research Agent, Lead Qualification Agent, Outreach Agent, CRM Update Agent, Follow-Up Agent) powered by Gemini / OpenAI LLMs.
- **7. Integration Layer:** External connectors managing real-time bidirectional synchronization with CRM Systems (Salesforce, HubSpot), Professional Networks (LinkedIn), Email APIs, Calendar Systems, and Video Conferencing Platforms (Zoom, Microsoft Teams).
- **8. Data Layer:** Enterprise persistence layer utilizing PostgreSQL, comprising dedicated logical repositories for the Lead Repository, Company Intelligence Database, Sales Activity Store, and Analytics Repository.

2.3 User Classes & Operational Characteristics

In compliance with IEEE Std 830-1998 Clause 5.2.3, the user classes participating in the platform are defined by their technical background, domain expertise, operational focus, and system privilege requirements.

User Class	Technical Background & Experience	Domain Skills & Operational Focus	System Access & UI Performance Requirements
Sales Representatives	General computer literacy; non-technical software background; minimal coding experience.	High-volume outreach, lead follow-up, client relationship execution, meeting reviews.	Zero-code interface simplicity; sub-5-second UI response (< 5 s); access restricted to assigned lead records and generation widgets.
Business Development Teams (BDRs)	Moderate technical literacy; familiar with CRM tools, data filtering, and lead enrichment.	In-depth prospect research, tech stack alignment analysis, cold account qualification.	Deep access to Company Profile Analysis, batch intelligence enrichment, expansion signals, and account target lists.
Sales Managers & RevOps Teams	Advanced technical capability; proficient in data analysis, BI tools, SQL queries, and system configuration.	Macro pipeline monitoring, conversion rate optimization, sales forecasting, CRM sync configuration.	Administrative access to executive dashboards, global CRM sync settings, audit logs, and RBAC configurations.

2.4 Database Schema & Data Model Architecture

The database layer is implemented in PostgreSQL following the relational model shown in Figure 2 and Table 2.4-A. The **Leads** entity is the central foreign-key target for all lead activity; **Sales_Analytics** is linked to **Users**.

Entity	Key Attributes	Foreign Key	Functional Context
Users	user_id (PK), name, email, role, department, created_at	Primary user record table.	Stores enterprise credentials, user role assignments, and organizational metadata.
Leads	lead_id (PK), company_name, industry, contact_name, email, phone, lead_status, created_at	Central FK target for all lead activity.	Stores core prospect demographics, company profile data, and current funnel stage.
Lead_Scores	score_id (PK), lead_id (FK), lead_score, conversion_probability, generated_at	lead_id → Leads.lead_id	Maintains historical AI predictive lead scores (0–100) and conversion probabilities.
Company_Insights	insight_id (PK), lead_id (FK), business_needs, opportunities, industry_analysis, generated_at	lead_id → Leads.lead_id	Holds AI-generated intelligence, tech stack compatibility data, and business signals.

Entity	Key Attributes	Foreign Key	Functional Context
Sales_Interactions	interaction_id (PK), lead_id (FK), interaction_type, summary, action_items, interaction_date	lead_id → Leads.lead_id	Stores meeting transcripts, conversation summaries, key discussion points, and action items.
Outreach_Campaigns	campaign_id (PK), lead_id (FK), email_subject, email_content, campaign_status, created_at	lead_id → Leads.lead_id	Stores AI-generated outreach templates, cold emails, and multi-channel campaign states.
CRM_Sync_Logs	sync_id (PK), lead_id (FK), crm_platform, sync_status, timestamp	lead_id → Leads.lead_id	Tracks real-time integration events, OAuth state, sync status, and API audit parameters.
Sales_Analytics	analytics_id (PK), user_id (FK), conversion_rate, pipeline_value, generated_at	user_id → Users.user_id	Stores aggregate sales pipeline performance metrics, conversion tracking, and pipeline valuation.

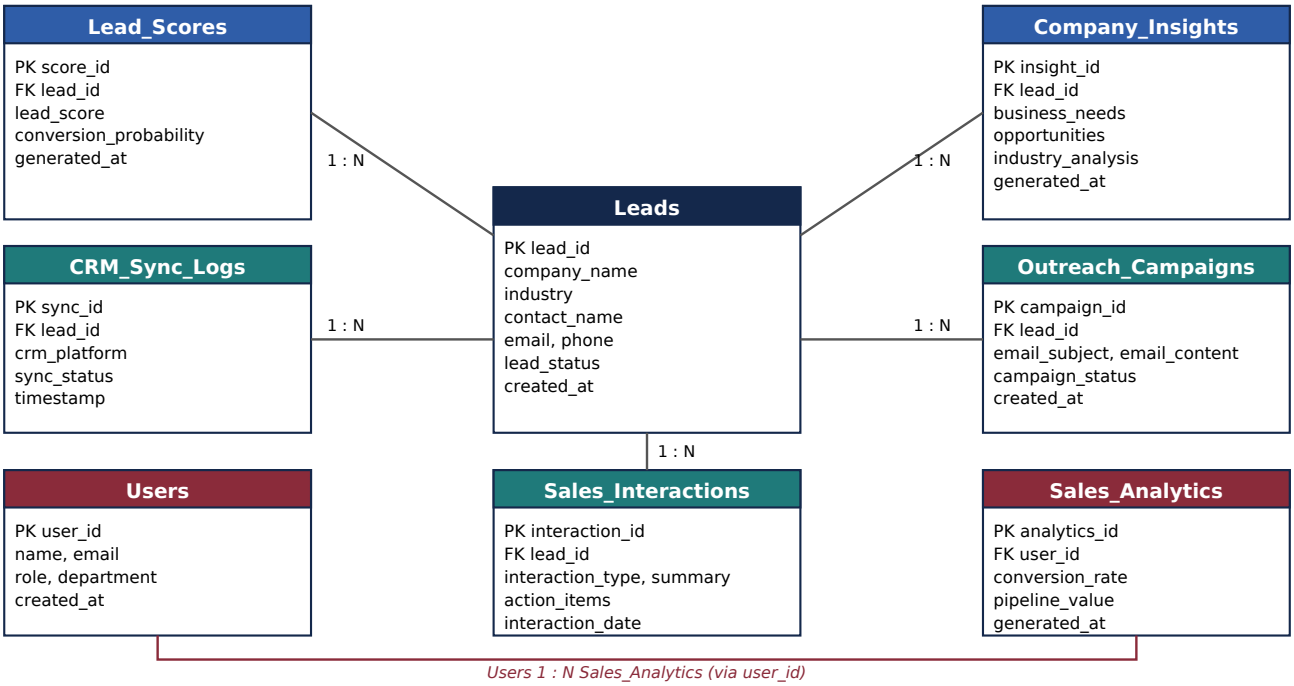


Figure 2 — Relational entity-relationship model (PostgreSQL). Teal entities relate to Leads; red entities relate to Users.

2.5 Operational Constraints & Dependencies

The SalesGenie AI runtime requires a dual-service execution process. The microservice REST backend is served by FastAPI using uvicorn, while the web frontend client is served concurrently using Streamlit. Third-party operational dependencies include external LLM API endpoint availability (Gemini / OpenAI), persistent network connectivity to third-party CRM webhooks (Salesforce, HubSpot), and PostgreSQL database connection availability.

Having defined the enterprise architecture, user classes, schema definitions, and operational dependencies, Section 3 details the core functional requirements across all platform modules.

3. Core Functional Requirements (Modules 1–6)

3.1 Strategic Functional Overview

All functional requirements are specified using Easy Approach to Requirements Syntax (EARS) statements — *When [event/stimulus], the system shall [response]* — to ensure unambiguous testability. Figure 3 shows the end-to-end system workflow pipeline.

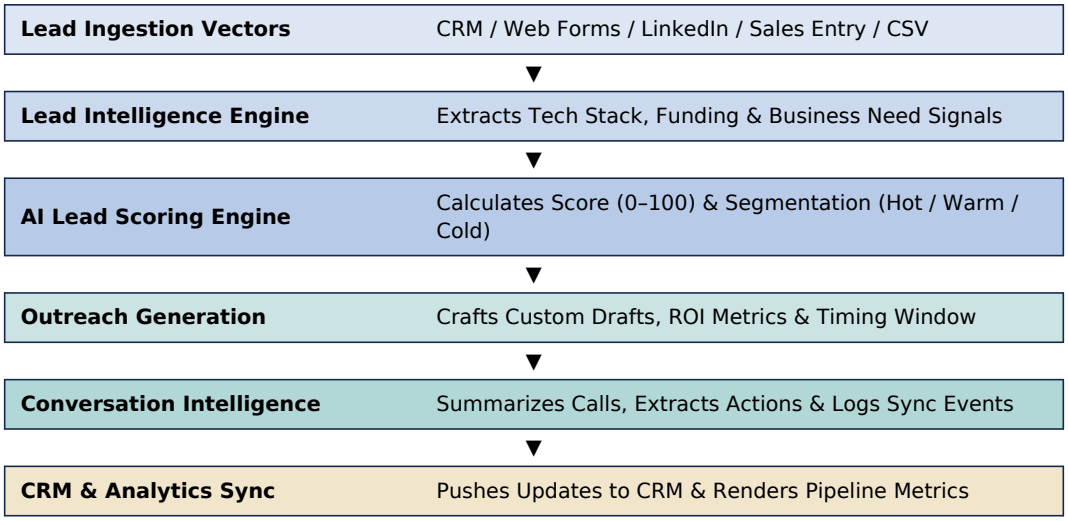


Figure 3 — End-to-end system workflow pipeline

Clarification Note on Architectural Artifacts: *The system workflow diagram references generated outputs produced by the platform execution pipeline. All generated outputs derive exclusively from the SalesGenie AI execution engine.*

3.2 Module 1: Lead Management & Prospect Database

Req. ID	Title	Requirement (EARS)
REQ-M1-01	Record Creation and Editing	When a user creates or modifies a prospect profile, the system shall validate, persist, and present the record within the PostgreSQL Lead Repository, displaying company name, primary contact details, and interaction history.
REQ-M1-02	Multi-Vector Data Ingestion	When prospect data is ingested via any of the 5 supported ingestion channels — CRM Leads, Website Forms, LinkedIn Prospects, Sales Team Entries, or CSV Uploads — the system shall parse, validate, and store comprehensive company demographic profiles including company size (e.g., 250–500 employees), annual revenue estimates (e.g., $45M–$60M), funding stage and capital raised (e.g., Series C, $28M), corporate location (e.g., San Francisco, CA), and active technology stack components (e.g., AWS, Python, React, Node.js, Kubernetes, PostgreSQL).
REQ-M1-03	Lifecycle Stage Tracking	When a prospect progresses through the sales funnel, the system shall update and display the current lead lifecycle stage across five predefined states: New Leads, Qualified, Proposal, Negotiation, and Closed Won.

3.3 Module 2: Lead Intelligence & Company Analysis

Req. ID	Title	Requirement (EARS)
REQ-M2-01	Automated Company Profiling	When a prospect record is selected or ingested, the Lead Intelligence Engine shall extract industry domain data, growth indicators, and technology stack compatibility to form an automated intelligence profile.
REQ-M2-02	Business Opportunity Identification	When evaluating prospect company profiles, the system shall analyze corporate attributes to detect and flag specific business transformation opportunities, explicitly identifying ERP Modernization Needs, Cloud Migration Opportunities, Process Automation Potential, and Digital Transformation Projects.
REQ-M2-03	Intelligence Summary Output	When profile intelligence processing completes, the system shall output an overall Qualification Score (e.g., 92) and structured qualitative insights, explicitly displaying High Growth Potential indicators (e.g., Series C funding signaling rapid expansion and tool budget), Tech Alignment metrics, and Decision Maker identification summaries.

3.4 Module 3: AI Outreach Generation

Req. ID	Title	Requirement (EARS)
REQ-M3-01	Personalized Email Generation	When an outreach generation request is submitted for a target prospect (e.g., sarah.johnson@techcorp.com), the AI Email Generator shall formulate a tailored cold email incorporating prospect context, funding milestones (e.g., congratulations on Series C funding), and explicit ROI benchmark references (e.g., “helped similar Series C companies like DataFlow reduce data processing time by 65% while cutting infrastructure costs by 40%”).
REQ-M3-02	Communication Strategy Customization	When generating outreach content, the system shall adjust message tone, value proposition structure, and channel strategy based on target vertical industry profiles and executive recipient persona roles.
REQ-M3-03	Optimal Timing and Channel Mix Recommendations	When outreach messages are generated, the system shall provide automated dispatch timing guidelines (e.g., “Send follow-up within 48 hours of initial email. Tuesday between 10–11 AM shows highest response rates; Optimal: Tuesday 10:00 AM”) and multi-channel engagement strategies (e.g., “After email, connect on LinkedIn within 24 hours referencing specific company achievements”).

3.5 Module 4: Lead Scoring & Recommendation Engine

Req. ID	Title	Requirement (EARS)
REQ-M4-01	Predictive Conversion Scoring	When prospect data is enriched or updated, the Lead Scoring Engine shall execute predictive evaluation models to calculate and output a numeric Lead Score (0–100) and an explicit Conversion Probability percentage (e.g., 78%).
REQ-M4-02	Explicit Scoring Factor Weighting	When computing predictive lead scores, the system shall derive points from dynamic scoring factors, explicitly displaying contributor weights including Company Growth (+25 points), Industry Match (+22 points), and Decision Maker title alignment.
REQ-M4-03	Automated Lead Segmentation	When lead scores are computed, the system shall categorize prospect profiles into operational priority buckets: Hot Leads, Warm Leads, and Cold Leads.
REQ-M4-04	Next-Best-Action Recommendations	When lead scores or user interactions update, the system shall generate prioritized next-best-action alerts (e.g., “Follow up with TechCorp: Sarah Johnson opened your proposal 3 times today. Schedule a call to discuss technical requirements [High Priority, 2 hours ago]” or “Send case study to DataFlow: Similar company in their industry achieved 65% efficiency improvement”).

3.6 Module 5: Conversation Intelligence & CRM Integration

Req. ID	Title	Requirement (EARS)
REQ-M5-01	Call and Meeting Summarization	When a meeting recording, transcript, or call notes are ingested (e.g., 45-minute discovery call with Sarah Johnson, CTO at TechCorp Solutions), the Conversation Intelligence Engine shall process the transcript to generate a structured Meeting Summary capturing explicit Key Discussion Points (e.g., “Data processing bottlenecks affecting customer experience”, “Need for real-time analytics and reporting capabilities”, “Budget approved for Q3 technology infrastructure upgrade”, “Competitive evaluation in progress with 2 other vendors”).
REQ-M5-02	Action Item Extraction with Assignment	When a meeting summary is generated, the system shall extract discrete action items, assigning ownership and due dates (e.g., “Alex Thompson: Send technical architecture document and integration guide — Due: Jun 10”; “Sarah Johnson: Schedule technical deep-dive with engineering team — Due: Jun 12”).
REQ-M5-03	Real-Time CRM Synchronization and Event Audit Logging	When customer interactions occur or deal stages change, the FastAPI Integration Layer shall execute bidirectional sync with external CRMs (Salesforce, HubSpot) and write real-time audit records to CRM_Sync_Logs, capturing exact event parameters (see Table 3.6-A).

Table 3.6-A — REQ-M5-03 audit log event types

Log Type	Example
Contact Sync Logs	Contact Synced [Added]: Sarah Johnson, CTO at TechCorp Solutions to Salesforce 2 min ago
Activity Logs	Activity Logged [Updated]: Initial outreach email sent and opened in HubSpot 15 min ago
Deal Stage Transition Logs	Deal Stage Updated [Moved]: Moved from ‘Prospecting’ to ‘Qualified’ in Salesforce 1 hour ago
Task Log Events	Task Created [New] in CRM

3.7 Module 6: Dashboard & Sales Analytics

Req. ID	Title	Requirement (EARS)
REQ-M6-01	Pipeline Metrics and Stage Visualizations	When accessed by authenticated users, the Streamlit Analytics Dashboard shall display interactive performance metrics rendered against baseline operational benchmarks (Tables 3.7-A and 3.7-B).
REQ-M6-02	Engagement Tracking	When active campaigns are executed, the system shall track and report customer engagement events, including email opens, link clicks, and reply rates.
REQ-M6-03	Business Intelligence Exporting	When requested by authorized RevOps users, the system shall compile historical interaction and conversion metrics into exportable business intelligence reports.

Table 3.7-A — Baseline dashboard metrics

Metric	Baseline Value	Change
Conversion Rate	24.8%	+3.2% from last month
Pipeline Value	$2.4M	+18% from last month
Average Response Time	2.4 hours	−45 min improvement
Average Sales Cycle	28 days	−3 days from last quarter

Table 3.7-B — Active pipeline funnel distribution (5 stages)

Stage	Leads	Example Accounts
New Leads	12	Acme Corp $120K, TechStart Inc $85K, Global Systems $65K
Qualified	8	DataFlow $210K, CloudNine $150K, TechCorp $180K
Proposal	5	InnovateAI $320K, WebScale $95K
Negotiation	3	FutureTech $275K, Quantum Corp $190K
Closed Won	6	NexGen $350K, AlphaTech $220K, PrimeSolutions $165K

With all functional modules defined, Section 4 outlines system performance metrics, IEEE 830 non-functional quality attributes, and the consolidated FMEA risk management matrix.

4. Key Non-Functional Requirements & Risk Management

4.1 Strategic Quality & Reliability Context

Enterprise adoption of agentic AI systems requires non-functional controls. High API latency, model hallucinations, or sync failures directly disrupt sales communication. Clear performance criteria and systematic failure mode analysis ensure platform reliability.

4.2 Performance & System Attributes

In compliance with IEEE Std 830-1998 Clause 5.3.6, the non-functional quality attributes of SalesGenie AI are defined with measurable targets and verification methods.

Quality Attribute	IEEE 830 Specification / Operational Target	Verification Method
Processing Latency	End-to-end lead profile analysis and personalized outreach generation completed in < 5 seconds.	Automated REST API load testing logging response latency under normal load.
Availability & Uptime	99.9% uptime for backend FastAPI microservices and database connection pools.	Synthetic health checks pinging /health endpoints at 60-second intervals.
Security & RBAC	Strict Role-Based Access Control enforced at FastAPI endpoints; encrypted API keys; OAuth 2.0 token isolation.	Automated security vulnerability scans and RBAC endpoint permission audits.
Maintainability & Scalability	Decoupled LangGraph multi-agent architecture allowing independent LLM engine swapping (Gemini/OpenAI) and DB scaling.	Static code analysis, microservice coupling audits, and database query profiling.
Portability	Deployment support across Linux (Ubuntu 22.04 LTS), macOS, and Windows environments via Docker containerization.	Containerized build validation across multi-platform CI/CD deployment pipelines.
Reliability & MTBF	Mean Time Between Failures (MTBF) ≥ 720 hours; maximum API error rate < 0.1% under peak burst load (100 req/sec).	Stress testing simulating concurrent peak user traffic and API payload bursts.

4.3 Risk Management & Failure Modes (FMEA Matrix)

Potential system failure modes are evaluated using Failure Modes and Effects Analysis (FMEA). Risk Priority Numbers are calculated as RPN = Severity (S) × Occurrence (O) × Detection (D) using standard 1–10 scoring scales.

Subsystem	Failure Mode	Cause / Mechanism	S	O	D	RPN	Technical Mitigation
AI Outreach Module	Hallucinated prospect facts in generated email content.	Unbounded LLM context window or high generation temperature (T > 0.7).	8	4	3	96	Enforce strict prompt grounding against PostgreSQL Lead Repository fields; execute zero-shot constraint check before rendering drafts in UI.
CRM Integration Layer	Synchronization failure or expired OAuth API tokens.	Third-party OAuth token expiration, rate limiting, or CRM network timeouts.	7	5	3	105	Implement automated retry queues with exponential backoff in FastAPI; log states in CRM_Sync_Logs and flag UI notification warnings.
Lead Scoring Engine	Prospect misclassification (high-value lead marked Cold).	Missing technology stack data or null corporate revenue fields during import.	6	4	4	96	Execute dynamic fallback attribute weighting when fields are null; trigger secondary agent research sub-workflows prior to score locking.

Having established non-functional quality attributes and risk controls, Section 5 details the implementation roadmap and milestone evaluation benchmarks.

5. Milestone Build Plan & Evaluation Metrics

5.1 Strategic Execution Context

Standards Compliance Note: *In accordance with IEEE Std 830-1998 Clause 4.8, formal project delivery schedules belong within a Software Project Management Plan (IEEE Std 1058). Section 5 is included here as an operational verification framework that bridges software requirements specifications to User Acceptance Testing (UAT) sign-off.*

5.2 Week-Wise Implementation Roadmap

Weeks 1–2	**Milestone 1 — Lead Management & Intelligence Engine** Schema Setup, Ingestion Vectors, Company Profiling Engine.
Weeks 3–4	**Milestone 2 — Outreach Generation & Lead Scoring Engine** AI Email Generator, Predictive Scoring, Strategy Recommendations.
Weeks 5–6	**Milestone 3 — CRM Integration & Conversation Intelligence** Call Summarization, Action Extraction, FastAPI Sync Endpoints.
Weeks 7–8	**Milestone 4 — Dashboard, Automation & Deployment** Streamlit Dashboard, Automated Widgets, E2E Optimization & Launch.

Figure 4 — 8-week implementation roadmap

Milestone 1 (Weeks 1–2) — Lead Management & Intelligence Engine

- Deploy PostgreSQL relational schema (Users, Leads, Company_Insights).
- Build lead management ingestion vectors (CRM Leads, Website Forms, LinkedIn Prospects, Sales Team Entries, CSV Uploads).
- Implement Company Research Engine and Lead Intelligence Generator for automated profile analysis.

Milestone 2 (Weeks 3–4) — Outreach Generation & Lead Scoring

- Build AI Email Generator with customized persona and industry tone adaptation.
- Develop predictive Lead Scoring Engine (0–100) and weighting factor calculations.
- Implement engagement strategy engine to output dispatch timing and channel mix recommendations.

Milestone 3 (Weeks 5–6) — CRM Integration & Conversation Intelligence

- Build Conversation Intelligence Engine for meeting transcript summarization and action item extraction.
- Develop FastAPI REST microservice endpoints for third-party integrations.
- Implement bidirectional CRM synchronization with Salesforce and HubSpot, logging events in CRM_Sync_Logs.

Milestone 4 (Weeks 7–8) — Dashboard & Automation

- Construct interactive Streamlit analytics dashboard rendering pipeline metrics, funnels, and recommendation widgets.
- Execute full integration testing, end-to-end optimization, and dual microservice deployment via uvicorn and Streamlit.

5.3 Milestone Evaluation Criteria & Acceptance Benchmarks

Milestone Phase	Operational Benchmark / Acceptance Criteria	Validation Method
Milestone 1 (Week 2)	Successfully ingest, parse, and analyze ≥ 95% of imported prospect records across all 5 ingestion vectors; produce accurate company profile summaries without field corruption.	Automated dataset ingestion tests and database schema integrity audits.
Milestone 2 (Week 4)	Achieve predictive lead scoring accuracy ≥ 85% against target validation datasets; generate personalized outreach content referencing valid ROI metrics across target test verticals.	Comparative scoring accuracy matrix and blind outreach quality review.
Milestone 3 (Week 6)	CRM integration functions correctly across supported workflows (Salesforce/HubSpot sync logs show success state); sales conversation summaries and action items generated successfully.	API integration test suites, OAuth renewal testing, and transcript summarization verification.
Milestone 4 (Week 8)	Analytics dashboard fully operational; end-to-end workflow validated; average lead analysis and outreach generation time < 5 seconds; overall user satisfaction score ≥ 85% during UAT evaluation.	Performance benchmark load testing, latency tracing, and UAT sign-off.

Acceptance: Satisfactory achievement and formal sign-off of all milestone evaluation criteria constitute full operational acceptance of the SalesGenie AI platform.